Alperen Tufan PELİT Student ID: 150720012

"On my honor, I have neither given nor received any unauthorized and/or inappropriate assistace for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to rash discipline issues"

Sign: A.T. Pelit

CamScanner ile tarandı

1-) $\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} = A$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & n & 0 & 1 \end{bmatrix}$, $\begin{bmatrix} -1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}$

$E \cdot A \simeq U$

$A = \underset{\text{Üst Üçg.}}{L} \cdot \underset{\text{Diagonal}}{D} \cdot \underset{\text{Alt Üçg.}}{U}$

$\alpha \begin{bmatrix} 0 & 1 & 0 & 1 \\ 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & \alpha \\ 0 & 0 & 0 & 0 \end{bmatrix} + \beta \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \gamma \begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & -9 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$+ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 1 & 14 & -6 & 2 \end{bmatrix}$

// A

Matrisedeki Belli Sabit Sayılar

$-8R_1 + R_2 = R_2$

$-8 \cdot 1 + 3\alpha = 0$

$\alpha = 8/3$

$m \cdot R_1 + R_3 = 0$

$m \cdot 1 + 2 = 0$

$m = -2$

1

CamScanner ile tarandı

c) Şimdi $n=2$ alacağız

$u = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$H = I - 2uu^T$

$H = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$A - \lambda I = 0$

$$\begin{vmatrix} -1-\lambda & 0 \\ 0 & 1-\lambda \end{vmatrix} = 0$$

$(-1-\lambda).(1-\lambda) = 0$

$\lambda_1 = 1$

$\lambda_2 = -1$

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3-) u is a unit vektor in R^n $n \times n$ matrix

$n=2$ $u = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $H = I - 2u.u^T$

$u^T.u = [1\ 0]\begin{bmatrix} 1 \\ 0 \end{bmatrix} = [1]$ ← Çarpım

a) $H^2 = I$

$(I - 2.u.u^T)^2 = I^2 - 2.I.u.u^T + 2.u.u^T.u.u^T$

$I - 2.u.u^T + 2.u.u^T$

I ✓

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b-) $H = I - 2u.u^T$ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}[1\ 0] = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

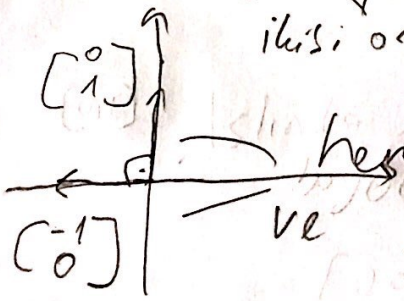
$u \quad u^T$

$H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$

$H = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

Orthonormal matrix demek içinde orthonormal vektörler bulunduruyor demektir. H matrixinde

$\begin{bmatrix} -1 \\ 0 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ ikisi orthonormal vektörlerdir.



hem açı 90° orthogonal ve birim vektörler old. için orthonormal oluyorlar

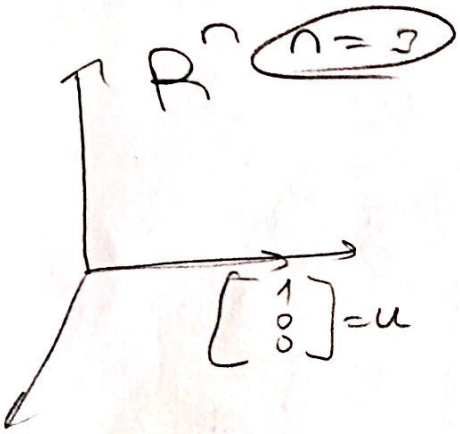
CamScanner ile tarandı

$$H = I - 2uu^T$$

u vector eigenvector

$$(H - \lambda I) \cdot u = 0$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



$$H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - 2 \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

$$H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$H = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{vmatrix} -1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{vmatrix} = 0$$

λ_1

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

λ_2

$$\begin{bmatrix} -2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(-1-\lambda)((1-\lambda)(1-\lambda)) = 0$$

$\lambda_1 = -1 \quad \lambda_2 = 1 \quad \lambda_3 = 1$

CamScanner ile tarandı

2) K dimensional space, $x-5y=0$

intersection with another geometry → Volume

a) Diğer uzay ile kesişimleri bir volume verebilmesi için, volume, en az 3 boyutlu olduğu için,

$x-5y+0z+0t=0$ ← $x-5y=0$ çözüm kümesi 4-D space'te bulunmalıdır.

eğer $x-5y=0$ 3-D space'te bulunsaydı bir plane, 2-D space'te bulunsaydı bir 1-D doğru çözüm kümesi olacaktı.

b-)

c-) $x+4y-z=6$ $x-5y+0z=0$

$x-5y+0z=0$

↓

$$\begin{bmatrix}1 & 4 & -1\\ 1 & -5 & 0\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}6\\ 0\end{bmatrix} \xrightarrow{-r_1+r_2=r_2}$$

$$\begin{bmatrix}1 & 4 & -1\\ 0 & -9 & 1\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}6\\ -6\end{bmatrix} \xrightarrow{-\frac{4}{9}R_2+R_1=R_1} \begin{bmatrix}1 & 0 & -13/9\\ 0 & -9 & 1\end{bmatrix}$$

next page →

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c) $x - 5y = 0 \quad x = 5y$

$x + 4y - z = 6$

$9y - z = 6$

Bunun 3-D içindeki çözüm kümesi 2-D boyutlu bir plane verir.

$$\begin{bmatrix} 0 & 9 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 6$$

free free

2 free olduğu için G.K. plane verir.

$$G.K = \alpha \begin{bmatrix} -9 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

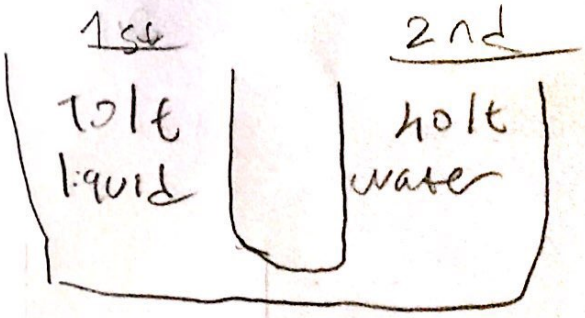
$(-F)$

Ama burada "0" yerine 6 olduğu için bu çözüm aslında geçerli olmaz. diğerini bulamadım.

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b) 2. soru b şıkkı

kesişimdeki düşen geometrinin volume olması için bu 2 matrixin, vektörlerinin en az 2 boyutta olması lazım



$10 - 40 = -30lt$

$$v_{n+1} = -2v_n + w_n$$

$$w_{n+1} = v_n - 2w_n$$

$$\int -v + w \rightarrow \frac{-v^2}{2} + \frac{w^2}{2}$$

$$\int v - w \rightarrow \frac{v^2}{2} - \frac{w^2}{2}$$

$$\int 2v - 2w \rightarrow v^2 - w^2$$

$$\int -v + w \rightarrow \frac{-v^2}{2} + \frac{w^2}{2}$$ ✓

Process'i açıklayan dif. eq. set 3. formüldür.

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b)

$$\begin{bmatrix} v_{k+1} \\ w_{k+1} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} v_k \\ w_k \end{bmatrix}$$

→ A, transition matrix

$$v_{k+1} = -2v_k + w_k$$

$$w_{k+1} = v_k - 2w_k$$

find eigenvalues, vectors

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} -2-\lambda & 1 \\ 1 & -2-\lambda \end{vmatrix} = (-2-\lambda)^2 - 1 = 0$$

$$4 + 4\lambda + \lambda^2 - 1 = 0$$

$$\lambda^2 + 4\lambda + 3 = 0$$

$$3 \quad 1$$

$$(\lambda+3)(\lambda+1) = 0$$

$\lambda_1 = -1$, $\lambda_2 = -3$ → eigen values

for λ_1

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} v_k \\ w_k \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ → eigenvector base

for λ_2

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} v_k \\ w_k \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ → eigenvector base

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$x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\lambda_1 = -1$ $\lambda_2 = -3$

$$\begin{bmatrix} a_n \\ b_n \end{bmatrix} = c_1 \cdot \lambda_1 x_1 + c_2 \lambda_2 \cdot x_2 = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot (-1)^n + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} (-3)^n$$

for $n=0$ $\begin{bmatrix} a_0 \\ b_0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$c_1 + c_2 = 1$$

$$c_1 - c_2 = 0$$

$$c_1 = 0{,}5$$
$$c_2 = 0{,}5$$

$$\begin{bmatrix} a_n \\ b_n \end{bmatrix} = 0{,}5 \cdot (-1)^k \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 0{,}5 \cdot (-3)^k \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

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